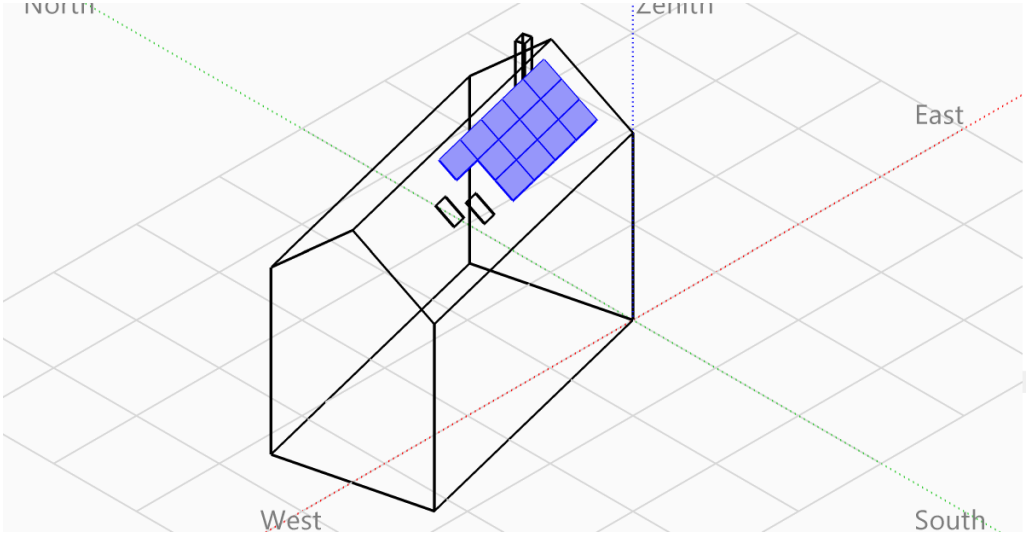
5.7 kWp rooftop PV with battery for a family house

From a real roof to a financed decision: load analysis, hand sizing of array, strings, inverter and battery, a 3D shading scene generated in AutoCAD, hourly PVsyst simulation of two cases (without and with storage) and a 25-year economic model in Excel.

Site — single-family house in the Nürnberg region, Bavaria

Load — 4-person household, 4,500 kWh/year

Tools — Hand calculation · AutoCAD + AutoLISP · PVsyst 8.1 · Excel



3D near-shading scene — house, chimney, skylights and 13 modules (PVsyst)

5.66 kWp	5.0 kW	6.56 kWh	1,132	39 → 70 %	8.2 yr
13 × SOLARWATT 435 Wp	SMA STP 5.0 Smart Energy	SMA Home Storage	kWh/kWp specific yield	autarky without → with battery	payback with battery

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Self-initiated concept study on a real building; dimensions from public map data. Not for construction.

A south-facing 40° roof, 14° east of south

Location & climate

Location	Nürnberg region, Bavaria
Altitude	≈ 340 m
Weather data	Meteonorm 9.0
Global horizontal (GHI)	1,154 kWh/m²
Mean ambient temp.	9.4 °C
Design T_{min} / T_{max}	−21.4 / 35.8 °C
Grid	400 V, 3-phase

Roof dimensions from public map data; exact location withheld for privacy. Irradiation: PVsyst / Meteonorm.

Building

Type	detached house
Footprint	16.48 × 8.14 m
Ridge height	13.19 m
Roof	gable, tilt 40°
Ridge axis	76° / 256°
PV roof face azimuth	−14° (SSE)
Slope length eave → ridge	5.31 m

Obstacles: chimney behind the ridge, two roof windows on the south face.

Design basis

Household	4 persons
Annual consumption	4,500 kWh
System type	on-grid, hybrid
Grid code	VDE-AR-N 4105
Single-phase limit	4.6 kVA
Tariff	EEG partial feed-in

Two values are **assumptions**, flagged throughout: roof tilt 40° (typical for German houses, not surveyed) and consumption 4,500 kWh (Stromspiegel 2025 for a 4-person house: 3,800–4,700 kWh). No utility bill was available.

Size by hand first, then let the simulation check it

01

Site & load data

Roof dimensions and orientation from public maps; consumption from German reference values.

`BayernAtlas · Google Earth · Stromspiegel 2025`

Output: design basis, pages 02–04

02

Hand sizing

Array size from daily energy, string window from temperature-corrected voltages, inverter ratio, battery from night demand.

`Datasheets · calculator`

Output: 13 modules, STP 5.0, 6.56 kWh

03

3D model

An AutoLISP script draws house, obstacles and modules from the measured dimensions and writes a Collada scene.

`AutoCAD · AutoLISP`

Output: shading scene (.dae)

04

PVsyst simulation

Hourly simulation with near shading, custom module and battery files, two variants: without and with storage.

`PVsyst 8.1 · Meteonorm 9.0`

Output: yield, losses, energy flows

05

Economics

25-year cash flow per case from the simulated energy flows; payback, NPV, IRR, LCOE and sensitivity.

`Excel model (formula-driven)`

Output: investment decision

Two cases, same PV array

Component	Without battery	With battery
PV array	13 × 435 Wp, 1 string	same
Inverter	STP 5.0 Smart Energy	same
Storage	none	Home Storage 6.5 (6.56 kWh)
Investment	7,350 €	10,960 €

The hybrid inverter is used in both cases, so the battery can be retrofitted later. The comparison isolates the value of storage alone.

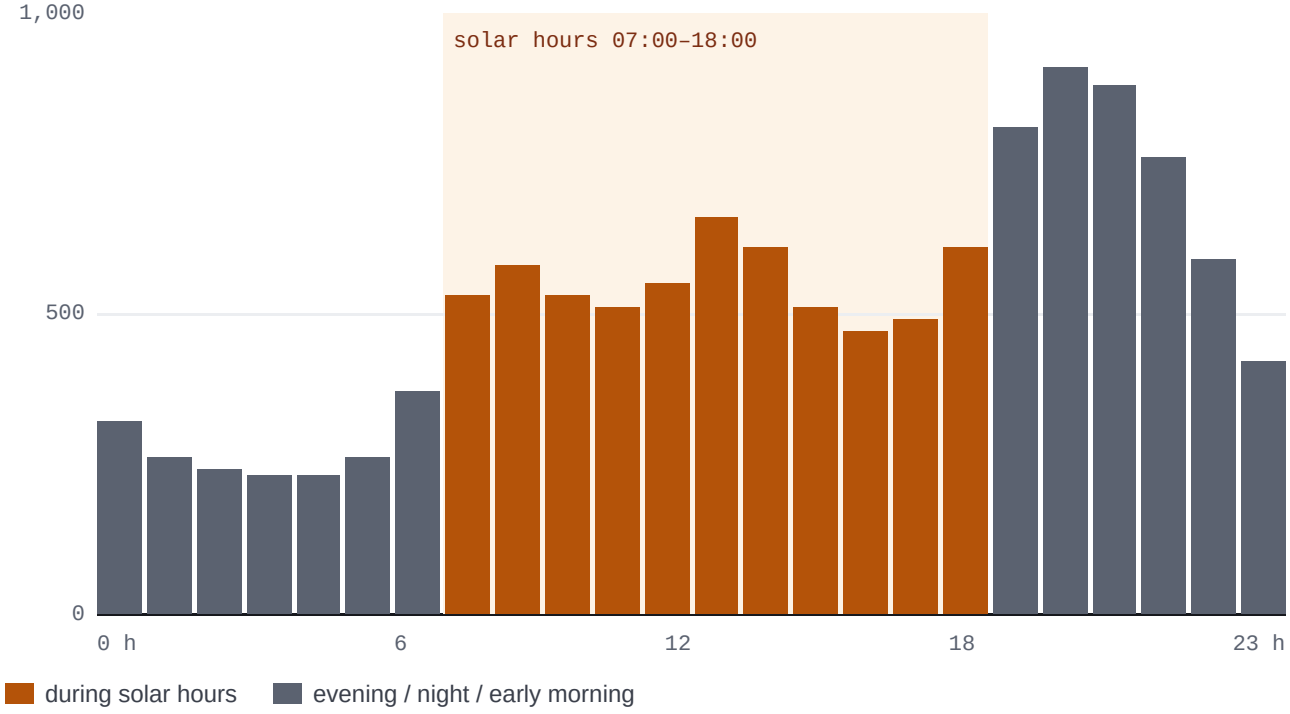
Questions the study answers

1. How many modules does a 4,500 kWh household need, and do they fit on the roof?
2. Which inverter and string length stay inside all voltage and current limits from −21 °C to +67 °C cell temperature?
3. How large should the battery be, and is it compatible with the inverter?
4. How much energy does the system produce, and how much of it does the house actually use?
5. Does the battery pay for itself?

Half of the daily demand falls outside solar hours

Hourly load, typical day (W)

12.33 kWh/day · constant over the year



How the profile was built

Annual demand from the Stromspiegel 2025 reference for a 4-person detached house; hourly shape modelled on the German standard load profile BDEW H0: low at night, a morning peak, a lunch peak and the main peak at 19–20 h.

Annual demand	4,500 kWh
Daily demand = 4,500 ÷ 365	12.33 kWh
Average power	514 W
Peak hour (19 h)	910 W
Energy 07:00–18:00	6.05 kWh (49 %)
Energy outside 07:00–18:00	**6.28 kWh (51 %)**

The 51 % night share is the number that sizes the battery (page 08). Without storage, PV can never cover it; this is why self-consumption stays low without a battery.

Limitation: one profile for all seasons. Real winter demand (lighting, circulation pump) is higher, so the winter grid share is likely underestimated.

13 modules cover the annual demand with margin

Energy-based sizing

E_{day} = 4,500 kWh ÷ 365 = 12.33 kWh

E_{req} = 12.33 × 1.3 (losses) = 16.03 kWh

P_{PV} = 16.03 kWh ÷ 3.0 h (PSH) = 5.34 kWp

N = 5,343 W ÷ 435 W = 12.28 → **13 modules**

P_{DC} = 13 × 435 Wp = **5.655 kWp**

Peak sun hours (PSH) of 3.0 h/day and a 1.3 loss factor are conservative design values. Rounded up, never down: the array must at least meet demand.

Cross-check with yield data

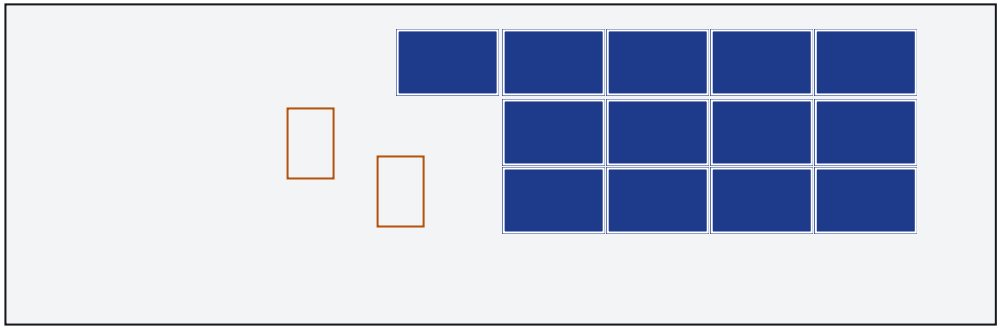
Source	kWh/kWp	Annual kWh	vs demand
Design rule (1 kWp → 1,000 kWh)	1,000	5,655	126 %
Global Solar Atlas, optimum tilt	1,126	6,365	141 %
PVsyst, this roof	**1,132**	**6,403**	**142 %**

Oversizing is intended: winter output is low, and the summer surplus charges the battery.

Module: SOLARWATT Panel classic AM 2.5 (435 Wp)

Pmax	Efficiency	Vmp / Imp	Voc / Isc
435 Wp	**22.3 %**	**32.6 V / 13.4 A**	**39.0 V / 13.9 A**
β Voc	γ Pmax	NMOT	Size
−0.25 %/K	**−0.31 %/K**	**45 °C**	**1722 × 1134 mm**

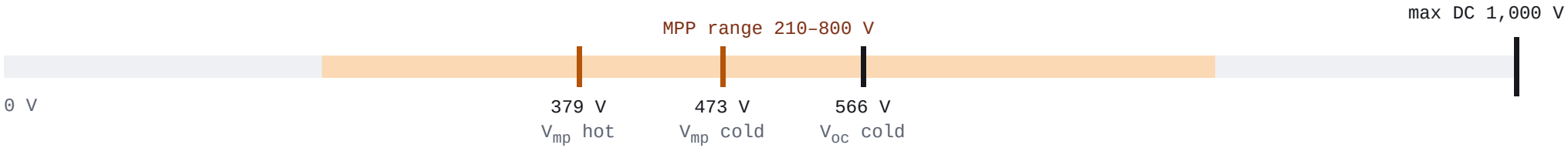
Roof fit, south face seen from the garden



W gable — 16.48 m · ridge at top, eave at bottom — E gable

Landscape, 4 columns × 3 rows + 1, packed against the east gable; 0.5 m edge and 0.4 m eave/ridge clearance; roof windows (orange) kept free. The roof face would hold 24 modules without obstacles.

One string of 13 stays inside the MPP window from −21 °C to 67 °C



String of 13 × SOLARWATT 435 Wp against the SMA STP 5.0-3SE-40 input limits

Cold morning: upper limit

$T_{cell} = T_{amb,min} = -21.4\ °C$

$V_{oc,cold} = 39.0 \times [1 - 0.0025 \times (-46.4)]$

$= 43.52\ V$

$N_{max} = 1{,}000 / 43.52 = 22.98 \rightarrow \mathbf{22}$

$V_{mp,cold} = 32.6 \times 1.116 = 36.38\ V$

Hot afternoon: lower limit

$T_{cell} = 35.8 + (45 - 20)/0.8 = 67.1\ °C$

$V_{mp,hot} = 32.6 \times [1 - 0.0025 \times 42.1] = 29.17\ V$

$N_{min} = 210 / 29.17 = 7.2 \rightarrow \mathbf{8}$

The upper limit is a safety limit, so it rounds down; the lower limit rounds up.

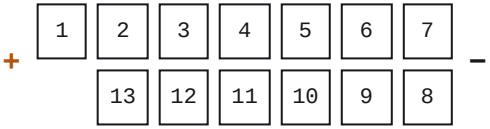
Checks for N = 13

Condition	String	Limit
$V_{oc,cold} \times 13$	565.8 V	< 1,000 V
$V_{mp,hot} \times 13$	379.2 V	> 210 V
$V_{mp,cold} \times 13$	473.0 V	< 800 V
I_{sc} per input	13.9 A	< 20 A
I_{mp} per input	13.4 A	> 12.5 A

I_{mp} slightly exceeds the 12.5 A input limit: the inverter caps the current on clear, cold days. This is safe (I_{sc} is within 20 A) and costs little energy, since the module rarely reaches STC in Bavaria.

A 7 + 6 split on both MPP inputs was rejected: 6 × 29.17 V = 175 V falls below 210 V on hot days.

THE STRING



13 modules in series on MPP input A

Power	5.655 kWp
V_{mpp} at STC	423.8 V
V_{mpp} range	379–473 V
V_{oc} at −21 °C	565.8 V
I_{mpp} / I_{sc}	13.4 / 13.9 A
Range allowed	**8 – 22**

A 5 kW three-phase hybrid inverter at DC/AC 1.13

AC power from the array

$P_{AC} = P_{DC} \div (1.1 \ldots 1.3)$

$5{,}655 \div 1.3 = 4{,}350\ W \ldots 5{,}655 \div 1.1 = 5{,}141\ W$

→ **5.0 kW**, DC/AC = 5,655 ÷ 5,000 = **1.13**

On-grid, the inverter is sized from the array, not from the load: the grid covers peaks and surges. A slightly smaller inverter is optimal because the array rarely reaches its STC rating.

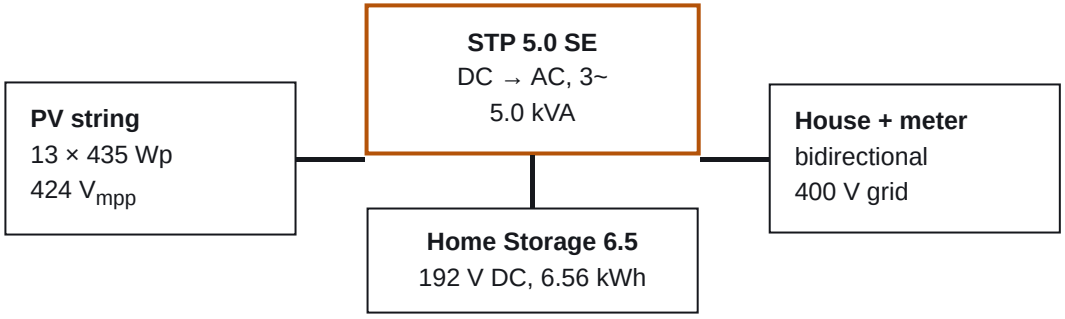
Options considered

Inverter	DC/AC	Verdict
SMA Sunny Boy 5.0 (1-phase)	1.13	no: 1-phase limit 4.6 kVA
SMA STP 4.0 Smart Energy	1.41	no: ratio above 1.3
SMA STP 6.0 Smart Energy	0.94	no: oversized, more cost
SMA STP 5.0 Smart Energy	**1.13**	**selected**

SMA STP5.0-3SE-40, key data

Rated AC power **5,000 VA**	Max. PV power **7,500 Wp**	Max. DC voltage **1,000 V**
MPP range **210–800 V**	Input current A / B **12.5 A / 12.5 A**	Isc max per input **20 A**
Battery voltage **150–600 V**	Battery current **30 A (U > 220 V)**	Efficiency max / euro **98.2 / 97.3 %**

System single-line



DC-coupled: the battery charges directly from the PV DC bus. The hybrid unit draws 44 W at night (datasheet), which PVsyst books as a 3 % loss.

6.56 kWh covers an average night and fits the inverter

1. Size from the night demand

E_{night} = 12.33 × 50 % = 6.2 kWh

$E_{PV,day}$ = 5,655 × 1,000 ÷ 365 = 15.5 kWh

Surplus = 15.5 − 6.2 = 9.3 kWh ≥ 6.2 ✓

C_{usable} = 6.2 ÷ η 0.945 = **6.52 kWh**

The surplus check matters: a battery is only useful if the array can fill it. In winter it cannot, which caps the benefit (page 12).

2. Rules of thumb (upper bound)

1 kWh per MWh/yr → 4.5 kWh

≤ 1 kWh per kWp → 5.7 kWh

Above this size each extra kWh is used on fewer days and the economics weaken.

3. Product choice

Only batteries on SMA's approved list communicate with the STP Smart Energy. The smallest approved SMA Home Storage is **6.5 (2 × 3.28 kWh modules)**. It matches the night demand and sits slightly above the rules of thumb; the excess is justified by the product range.

Compatibility checks

Check	Value	Result
Approved for STP-SE	HS 6.5–16.4	OK
Voltage in 150–600 V	180–216 V	OK
Usable ≥ required	6.56 ≥ 6.52 kWh	OK
Current = min(36, 30)	30 A	OK
Power = 192 V × 30 A	5.76 kW	OK
C-rate = 5.76 ÷ 6.56	0.88 C	OK
U_{bat} > 220 V for full 30 A	192 V	note

Below 220 V SMA does not guarantee the full 30 A. Sensitivity: even at 20 A the battery delivers 192 × 20 = 3.8 kW, above the 0.91 kW peak hour of the profile and a 3 kW evening peak. The decision is robust.

SMA HOME STORAGE 6.5

HS-BM-3.28 · 96 V

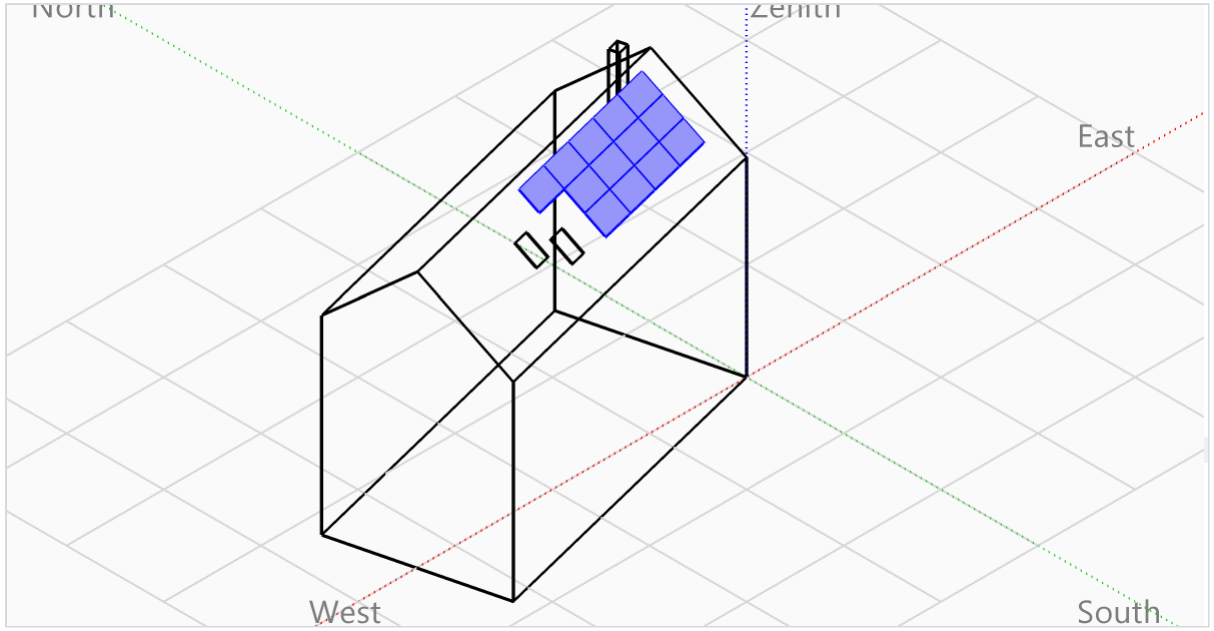
HS-BM-3.28 · 96 V

2 modules in series → 192 V

Usable energy	6.56 kWh
Chemistry	LFP
Max. current	36 A
Efficiency	94.5 %
Cycle life	8,000
Warranty	10 years
Expandable to	**16.4 kWh**

Modelled in PVsyst as a custom battery file (no SMA entry in the library). SOC window 5–98 % → 6.10 kWh used.

A scripted 3D scene: near-shading loss of only 0.17 %



PVsyst Construction/Perspective after import — 13 PV fields, 25.4 m², tilt 40° / azimuth −14°

−0.17 %	−2.78 %	+16.2 %	−0.6 %
near shading	IAM (glass)	tilt gain vs GHI	vs optimal orientation

From measurements to scene in one command

1. **Parameters** — footprint, ridge height, tilt, module size, edge clearances and obstacle positions at the top of an AutoLISP file.
2. **Geometry** — house extruded from the gable profile; eave height derived: 13.19 − 4.07 × tan 40° = 9.77 m.
3. **Module placement** — modules drawn flat, then transformed onto the 40° plane 10 cm above the tiles; obstacles skipped with 0.2 m clearance.
4. **Orientation** — model rotated 104° so the PV face points to azimuth −14°.
5. **Export** — a Collada (.dae) file written directly by the script, PV top surface as its own material, imported into PVsyst without Blender or MeshLab.

Debugging step: the first export also carried module side faces, which PVsyst flagged as objects touching the PV fields. Exporting the PV surface only removed the warning.

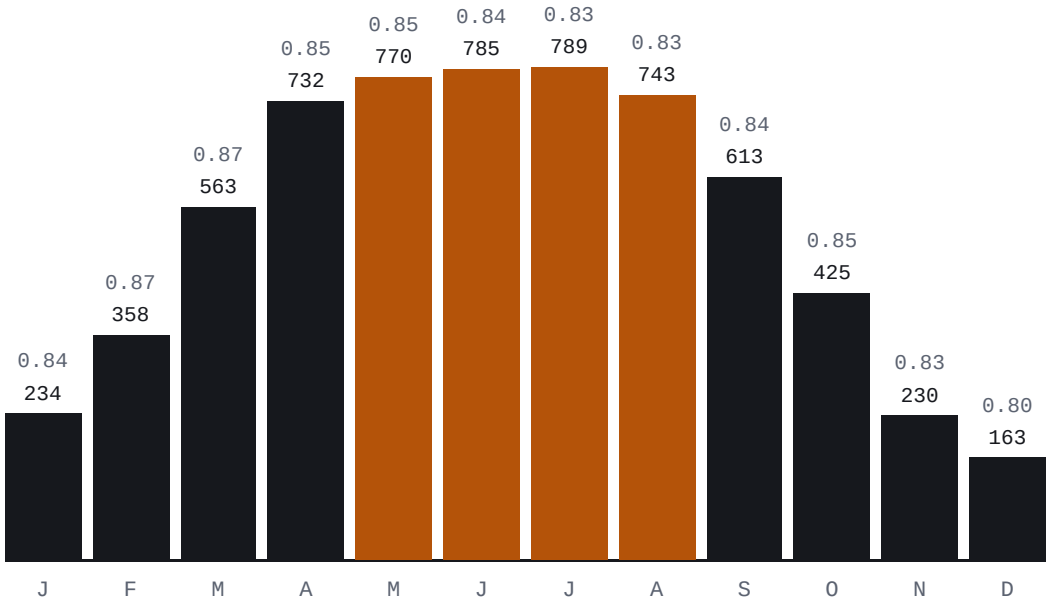
Why the loss is low: the chimney sits north of the ridge and the roof windows are east of the array, so neither casts a shadow on it while the sun is in front of the plane.

6,403 kWh per year at a performance ratio of 84 %

6,403 kWh	1,132	84.4 %	1,341	6,060 kWh
AC energy, year 1	kWh/kWp specific yield	performance ratio	kWh/m² on the plane	P90 (variability 4.2 %)

Monthly AC energy

kWh · PR above each bar



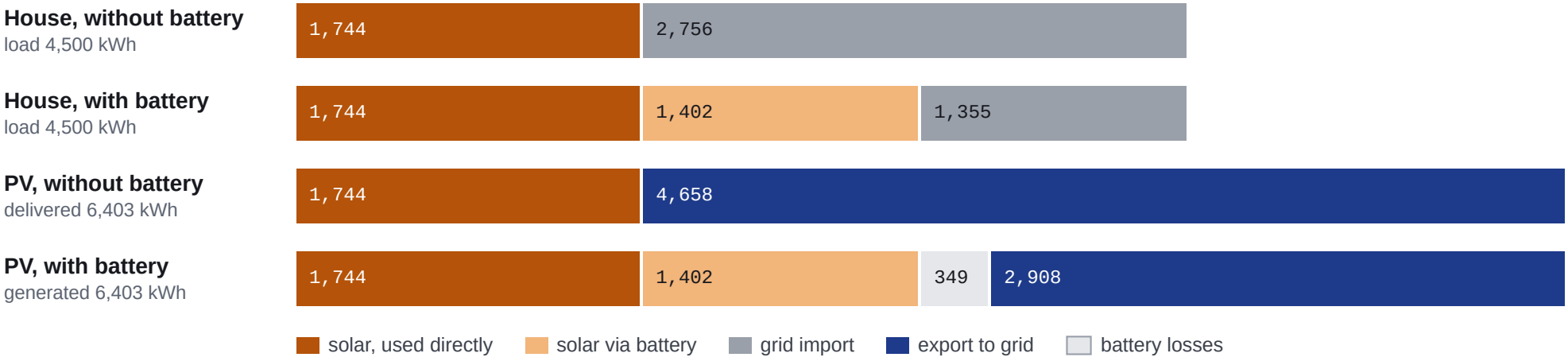
July delivers 4.9 × the December yield. December PR dips to 0.80: the fixed 44 W night draw and low-light losses weigh more on a small output.

Where the energy goes (loss diagram)

Global horizontal irradiation	**1,154 kWh/m²**
Transposition to 40° / −14°	+16.2 %
Near shading	−0.17 %
IAM (angle of incidence)	−2.78 %
Array nominal energy at STC	**7,359 kWh**
Low irradiance	−1.11 %
Module temperature	−3.51 %
Module quality (+5 W tolerance)	+1.25 %
Mismatch + DC wiring	−3.09 %
Inverter efficiency	−3.91 %
Inverter clipping (DC/AC 1.13)	−0.19 %
Inverter night consumption	−3.00 %
AC energy delivered	**6,403 kWh**

Thermal model: semi-integrated, U_c = 20 W/m²K (modules 10 cm above tiles). Night consumption 44 W of the hybrid inverter = 3 %.

The battery moves 1,400 kWh a year from the grid to the sun



Key indicators (PVsyst, year 1)

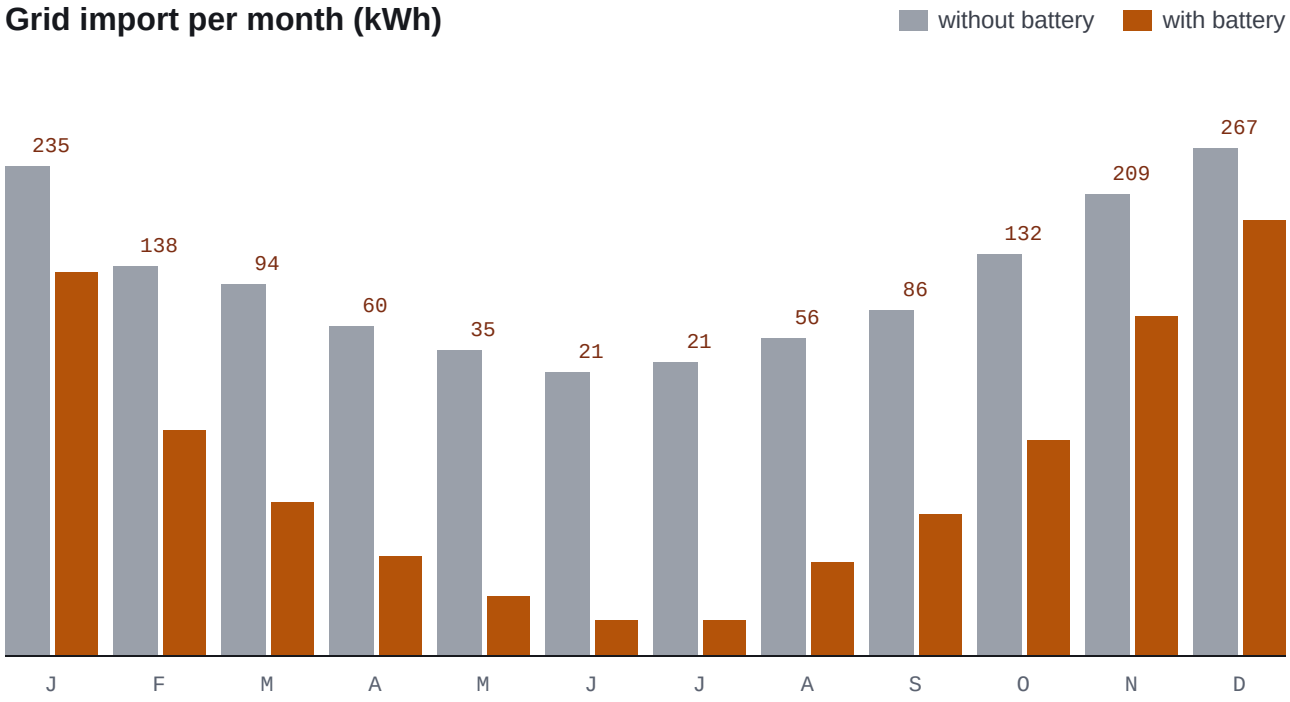
kWh / year	Without battery	With battery	Change
Solar used in the house	1,744	3,146	+1,402
Grid import	2,756	1,355	−1,401
Export	4,658	2,908	−1,751
Autarky = solar used ÷ load	38.8 %	**69.9 %**	+31 pts
Self-consumption = used ÷ delivered	27.2 %	52.0 %	+25 pts

Reading the balance

1. Midday surplus that went to the grid is now stored: export falls by **1,751 kWh**.
2. The battery returns **1,402 kWh** in the evening and night, replacing grid purchases one to one.
3. The difference, **≈ 350 kWh**, is lost in charging, storage and discharging (≈ 80 % round trip for the full DC→AC path).
4. About 1,402 ÷ 6.1 ≈ **230 equivalent full cycles** per year, far below SMA's 8,000-cycle rating.

Summer is almost grid-free; winter is where the remaining 30 % comes from

Grid import per month (kWh)



Monthly autarky with battery

J	F	M	A	M	J	J	A	S	O	N	D
38%	60%	75%	84%	91%	94%	95%	85%	77%	66%	44%	30%

What the months show

Month	PV kWh	Import w/o	Import with
July	789	180	**21**
April	732	202	60
October	425	247	132
December	163	311	**267**

From May to August the battery is full almost every evening and the house runs 85–95 % on solar. In December the array produces 163 kWh against a 382 kWh load: there is hardly any surplus to store, so autarky only rises from 19 % to 30 %.

Consequence for design: a larger battery would not fix winter. Only more winter yield (more modules, steeper tilt) or a different load (heat pump with PV-optimised control) would.

The hand design held; the energy estimate was 13 % conservative

Quantity	Hand calculation	PVsyst	Deviation	Why
Specific yield	1,000 kWh/kWp	1,132 kWh/kWp	+13 %	design rule ignores the 16 % tilt gain
Peak sun hours (after losses)	3.0 h/day	3.10 h/day	+3 %	1,132 ÷ 365
Annual AC energy	5,655 kWh	6,403 kWh	+13 %	follows from yield
Performance ratio	0.80 (assumed)	0.844	+0.04	low shading, cool climate
DC/AC ratio	1.13	1.13	0	
String V_{oc} at −21 °C	565.8 V	568 V	+0.4 %	PVsyst model β_{oc} −0.27 %/K
String Vmpp at 50 °C	397 V	389 V	−2.0 %	PVsyst uses γ Vmp −0.32 %/K
Inverter clipping	"small"	0.19 %		ratio 1.13, Bavarian irradiance
Battery energy to house	6.2 kWh × 365 = 2,250	1,402 kWh	−38 %	no surplus in winter months
Autarky with battery	60–65 % (estimate)	69.9 %	+5 pts	evening peak well covered in summer

Electrical design: confirmed

Voltages agree within 2 %, DC/AC is identical and clipping is negligible. No change to strings or inverter needed.

Yield: hand rule is conservative

The 1,000 kWh/kWp rule underestimates this roof by 13 %; Global Solar Atlas (1,126) and PVsyst (1,132) agree within 1 %.

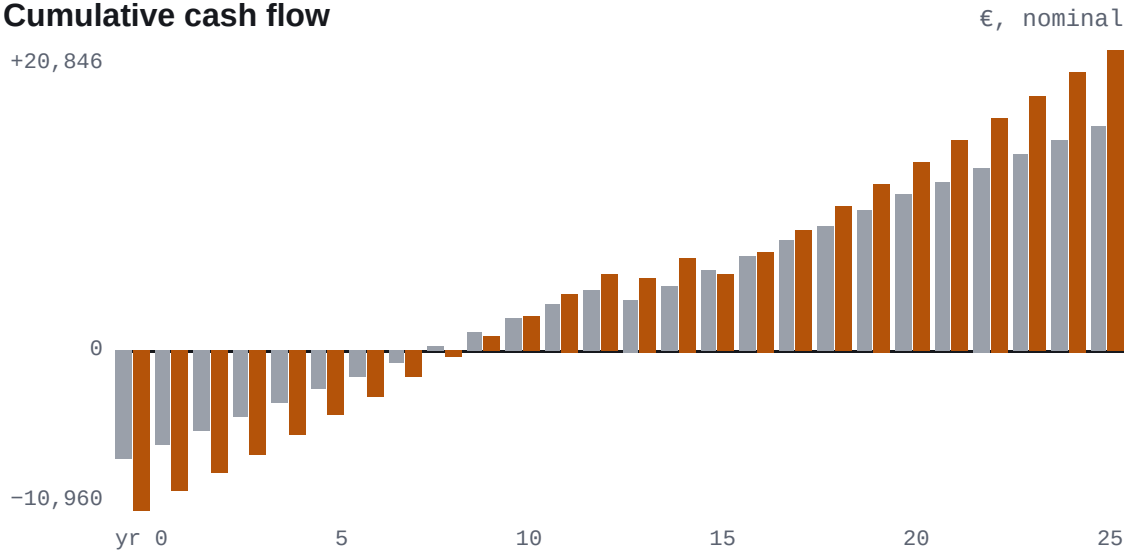
Battery: limited by winter

The daily night-energy method assumes a full battery every day. The hourly simulation shows it delivers 62 % of that potential, because winter surplus is small.

Both cases pay back in 8 years; the battery adds 2,400 € of value

25 years, nominal €	Investment	Net cash, yr 1	Payback	NPV at 3 %	IRR	LCOE	Cumulative yr 25
Without battery	7,352	931	7.7 yr	+8,566	11.7 %	9.6 ct	+15,549
With battery	10,960	1,278	8.2 yr	**+11,013**	10.6 %	16.2 ct	**+20,846**
Battery alone (difference)	3,608	348	9.5 yr	+2,447	8.2 %	–	+5,297

Cumulative cash flow



Dips in year 13 (inverter replacement, both cases) and year 15 (battery replacement). After year 20 the EEG tariff ends and export earns only the 4 ct market value.

Assumptions (net prices, 2026)

PV system, turnkey	1,300 €/kWp	42watt 07/2026
Battery, installed	550 €/kWh	42watt 07/2026
Grid price	37 ct, +2 %/yr	BDEW 08/2026
Feed-in (EEG, ≤ 10 kWp)	7.7 ct, 20 yr	rate from 08/2026
Export after year 20	4.0 ct	assumption
O&M incl. insurance	1 %/yr, +2 %	assumption
Inverter replacement, yr 13	1,800 €	assumption
Battery replacement, yr 15	400 €/kWh	assumption
Degradation	0.5 %/yr	assumption
Discount rate / lifetime	3 % / 25 yr	household
Energy flows	PVsyst yr 1	pages 10–11

Every kWh used at home saves 37 ct; every kWh exported earns 7.7 ct. The battery's value is the spread between the two, on about 1,400 kWh a year.

Year by year with battery: the investment is back in year 9

n	Year	Self-used kWh	Export kWh	Savings €	Export €	Costs €	Net €	Cumulative €
0	2026	–	–	–	–	–	**−10,960**	−10,960
1	2027	3,146	2,908	1,164	224	110	**1,278**	−9,681
2	2028	3,130	2,893	1,181	223	112	**1,292**	−8,389
3	2029	3,114	2,879	1,199	222	114	**1,307**	−7,082
4	2030	3,099	2,864	1,217	221	116	**1,321**	−5,761
5	2031	3,083	2,850	1,235	219	119	**1,336**	−4,426
6	2032	3,068	2,836	1,253	218	121	**1,351**	−3,075
7	2033	3,053	2,821	1,272	217	123	**1,366**	−1,709
8	2034	3,037	2,807	1,291	216	126	**1,381**	−328
9	2035	3,022	2,793	1,310	215	128	**1,397**	1,069
10	2036	3,007	2,779	1,330	214	131	**1,413**	2,481
11	2037	2,992	2,765	1,349	213	134	**1,429**	3,910
12	2038	2,977	2,752	1,370	212	136	**1,445**	5,355
13	2039	2,962	2,738	1,390	211	1,939	**−338**	5,017
14	2040	2,947	2,724	1,411	210	142	**1,479**	6,496
15	2041	2,933	2,711	1,432	209	2,769	**−1,128**	5,368
16	2042	2,918	2,697	1,453	208	148	**1,513**	6,881
17	2043	2,903	2,684	1,475	207	150	**1,531**	8,412
18	2044	2,889	2,670	1,497	206	153	**1,549**	9,961
19	2045	2,874	2,657	1,519	205	157	**1,567**	11,528
20	2046	2,860	2,643	1,542	204	160	**1,585**	13,113
21	2047	2,846	2,630	1,565	105	163	**1,507**	14,620
22	2048	2,831	2,617	1,588	105	166	**1,526**	16,147
23	2049	2,817	2,604	1,612	104	169	**1,546**	17,693
24	2050	2,803	2,591	1,636	104	173	**1,566**	19,259
25	2051	2,789	2,578	1,660	103	176	**1,587**	20,846
25 years		**74,102**	**68,491**	**34,948**	**4,792**	**7,934**		**20,846**

How each column is calculated

1. **Self-used, export** = PVsyst year-1 value × (1 − 0.5 %)$^{n-1}$ (3,146 / 2,908 kWh)
2. **Savings** = self-used kWh × 0.37 € × 1.02^{n-1}
3. **Export** = exported kWh × 0.077 € (years 1–20), × 0.04 € after
4. **Costs** = 1 % × 10,960 € × 1.02^{n-1}, plus 1,800 € inverter (yr 13) and 6.56 × 400 € battery (yr 15)
5. **Net** = savings + export − costs; year 0 = −10,960 € investment

How the results follow

Payback = 8 + 328 ÷ 1,397 (cumulative crosses 0 in year 9)	**8.2 yr**
NPV = Σ Net$_n$ ÷ 1.03^n	**11,013 €**
IRR — discount rate at which NPV = 0	**10.6 %**
LCOE = Σ discounted costs ÷ Σ discounted kWh	**16.2 ct**

payback year replacement year

The case without battery uses the same model with 1,744 / 4,658 kWh, 7,352 € CAPEX and no battery replacement. Both sheets are in the Excel calculator.

All values from **PV_Residential_Economics.xlsx**; every blue input cell can be changed and the model recalculates.

The battery stays worth it in every tested case

One input changed at a time

Case	Payback w/o	Payback with	NPV w/o	NPV with	Battery adds (NPV)
Base case	7.7	8.2	8,566	11,013	+2,447
Grid price 32 ct	8.5	9.4	6,782	7,797	+1,015
Grid price 42 ct	7.1	7.4	10,349	14,229	+3,880
Battery 400 €/kWh	7.7	7.5	8,566	12,210	+3,644
Battery 700 €/kWh	7.7	9.0	8,566	9,816	+1,250
PV 1,100 €/kWp	6.5	7.4	9,941	12,389	+2,448
PV 1,500 €/kWp	9.0	9.1	7,190	9,637	+2,447
Feed-in 5.2 ct	8.7	8.7	6,905	9,976	+3,071

Payback in years, NPV in € at 3 % over 25 years. Each row re-runs the full Excel model with one input changed.

RECOMMENDATION

Install PV with the 6.56 kWh battery

Highest NPV	11,013 €
Grid purchase	−51 %
Net bill, year 1	1,665 → 277 €
Payback	8.2 years

PV alone pays back half a year faster, but the battery is a positive investment on its own (IRR 8.2 %, above the 3 % discount rate) and protects against falling feed-in tariffs. The hybrid inverter keeps the option to add modules up to 16.4 kWh later.

Grid price drives everything

±5 ct moves the battery's added value between 1,000 € and 3,900 €.

Lower feed-in favours storage

At 5.2 ct (2027 draft) PV-only loses 1,660 €, the battery case only 1,040 €.

PV cost doesn't change the choice

It shifts both cases equally; the battery's added value stays 2,450 €.

What a real project would verify next

Topic	Current state in this study	Next step	Impact
Consumption	4,500 kWh from reference values, one daily profile for all seasons	Utility bill and smart-meter data (15-min) from the household	high
Roof survey	Tilt 40° assumed; obstacle positions from aerial images	On-site measurement or drone survey; check rafters and tile type	medium
Battery ageing	Generic ageing curve in PVsyst; replacement assumed in year 15	Enter SMA cycle data (8,000 cycles) and re-run	medium
Prices	Market averages for 2026, not a quote	Two installer quotes; check KfW and local subsidies	medium
Feed-in rules	EEG 7.7 ct for 20 years; 2027 reform pending	Re-check with the final law; consider dynamic tariff	medium
Imp vs input limit	Imp 13.4 A above 12.5 A per input; PVsyst shows 0 % loss	Confirm clipping with SMA Sunny Design	low
Electrical details	No cable sizing or protection design yet	DC/AC cable cross-sections, SPD, meter cabinet	low

Deliverables of this study

Hand calculation (pages 04–08) · AutoLISP 3D scene generator · PVsyst reports for both cases · Excel economic calculator with 25-year cash flows and sensitivity

Skills shown

Load analysis · array, string, inverter and battery sizing · grid-code constraints · 3D modelling for shading · PVsyst simulation and loss analysis · self-consumption and storage analysis · investment appraisal (payback, NPV, IRR, LCOE)

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